

Linear Block Code

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Linear Block Code

Definition of Linear Block Code

- A block code is said to be a linear block code if its codewords satisfy the condition that the sum (modulo-2 addition) of any two codewords results in another valid codeword.
 - Example: $C_p = C_i + C_k$

Properties of Linear Block Code

1. The all-zero word $[0, 0, 0, \dots, 0]$ is always a valid codeword in a Linear Block Code.
2. Minimum Hamming Distance of the Code $d_{\min} = W_{\min}$.
3. It can detect up to $(d_{\min} - 1)$ Errors.
4. It can correct up to $\lfloor \frac{d_{\min} - 1}{2} \rfloor$ Errors.

Hamming Code (7,4)						
D	D3	D2	P3	D1	P2	P1
4						
0	0	0	0	0	0	0
0	0	0	0	1	1	1
0	0	1	1	0	0	1
0	0	1	1	1	1	0
0	1	0	1	0	1	0
0	1	0	1	1	0	1
0	1	1	0	0	1	1
0	1	1	0	1	0	0
1	0	0	1	0	1	1
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	0	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	0
1	1	1	1	0	0	0
1	1	1	1	1	1	1

Checking whether Hamming Code is a Linear Block Code

1. For Linear Block Code, $C_p = C_i + C_j$.
2. The all-zero word $[0, 0, 0, \dots, 0]$ is always a valid codeword in a Linear Block Code.
3. Minimum Hamming Distance of the Code $d_{\min} = W_{\min}$.
4. It can detect up to $(d_{\min} - 1)$ Errors.
5. It can correct up to $\lfloor \frac{d_{\min} - 1}{2} \rfloor$ Errors.

Example of Linear Block Code

- ❑ **Example 1: Show that the (4, 3) Even-Parity Code is linear and the (4, 3) Odd-Parity Code is not linear.**

(4, 3) Even Parity Block Code

Data			Parity
D3	D2	D1	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

(4, 3) Odd Parity Block Code

Data			Parity
D3	D2	D1	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

- **$C_p = C_i + C_k$, It is followed in (4, 3) Even Parity Block Code, but it is not followed in (4, 3) Odd Parity Block Code.**

Linear Block Code Solved

□ Example 1: For a (6, 3) code, the Generator Matrix G

is given by:

$$\begin{matrix} (n,k) \\ \swarrow \quad \searrow \\ n=6 \quad k=3 \end{matrix} \Rightarrow G = \left[\begin{array}{c|ccc|c} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right] = [I_k : P]$$

□

- I. All Corresponding Code Vectors
- II. Minimum Hamming Distance
- III. Error Detection and Error Correction Capability
- IV. Parity Check Matrix
- V. Find error if received code is [1 0 0 0 1

→ Identity matrix I_k

→ Parity matrix P

$$I_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\leadsto \text{Codeword } [c] = [i][k] = [i : p_c]$$

$\leadsto$ Parity Codeword Matrix

$$\Rightarrow [p_c] = [i][P]$$

$$\Rightarrow [p_0 \ p_1 \ p_2] = [i_0 \ i_1 \ i_2] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow p_0 = i_0 \oplus i_2$$

$$p_1 = i_1 \oplus i_2$$

$$p_2 = i_0 \oplus i_1$$

c	i ₀	i ₁	i ₂	p ₀	p ₁	p ₂	w
c ₁	0	0	0	0	0	0	1
c ₂	0	0	1	1	1	0	3
c ₃	0	1	0	0	1	1	3
c ₄	0	1	1	1	0	1	4
c ₅	1	0	0	1	0	1	3
c ₆	1	0	1	0	1	1	4
c ₇	1	1	0	1	1	0	4
c ₈	1	1	1	0	0	0	3

$\leadsto$ Minimum Hamming Distance $d_{\min} = w_{\min} = 3$

$\leadsto$ Error detection $\leq d_{\min} - 1$
 ≤ 2 bits

$\leadsto$ Error correction $\leq \left(\frac{d_{\min} - 1}{2} \right)$
 ≤ 1 bits

→ Parity check matrix $[H]$

$$\Rightarrow [H] = [P^T : I_{n-k}] = [P^T : I_3]$$

→ Parity matrix P

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow P^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

→ Syndrome matrix $[s]$

$$\Rightarrow [s] = [s][H^T]$$

$$= [1 \ 0 \ 0 \ 0 \ 1 \ 1] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

at 3rd row

$$= [1 \ 1 \ 0]$$

→ So error is at 3rd bit.

$$\Rightarrow e = [0 \ 0 \ 1 \ 0 \ 0 \ 0]$$

$$f = [1 \ 0 \ 0 \ 0 \ 1 \ 1]$$

$$\Rightarrow x = e + f$$

$$= [1 \ 0 \ 1 \ 0 \ 1 \ 1]$$

Linear Block Code Solved

□ Example 2: Determine the set of codewords for a (6, 3) code, the Generator Matrix G is given by:

Examples

$$\Rightarrow G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} = [I_k : P]$$

$$\leadsto (6, 3) = (n, k)$$

$$- n = 6$$

$$- k = 3$$

$\leadsto$ Identity Matrix

$$I_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\leadsto$ Parity Matrix

$$P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\leadsto \text{Codeword } [c] = [i][G] = [i : P]$$

⇒ Parity Codeword matrix

$$\Rightarrow [P_c] = [i][P]$$

$$\Rightarrow [P_0 \ P_1 \ P_2] = [i_0 \ i_1 \ i_2] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow P_0 = i_1 \oplus i_2$$

$$P_1 = i_0 \oplus i_2$$

$$P_2 = i_0 \oplus i_1$$

C	i_0	i_1	i_2	P_0	P_1	P_2
C_1	0	0	0	0	0	0
C_2	0	0	1	1	1	0
C_3	0	1	0	1	0	1
C_4	0	1	1	0	1	1
C_5	1	0	0	0	1	1
C_6	1	0	1	1	0	1
C_7	1	1	0	1	1	0
C_8	1	1	1	0	0	0

□ Example 3: The (5, 3) linear code has the generator

Matrix G : $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} = [i_k : p]$

$\Rightarrow G$

$=$

▪ Determine Systematic form of G .

▪ Generate Codeword for information [0 1 1] with systematic G and Non-systematic G .

$\rightarrow (5, 3) = (n, k)$

$n = 5$

$k = 3$

$\rightarrow G = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Add}} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{\text{Add}} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

$R_3 = R_2 \oplus R_3$

$R_1 = R_1 \oplus R_3$

$[G] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} = [i_k : p]$

$$\sim C = [i][k]$$

$$= [0 \ 1 \ 1] \left| \begin{array}{ccccc} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{array} \right|$$

$$= [0 \ 0 \ 1 \ 1 \ 1]_{\text{Non-Systematic}}$$

$$\sim C = [i][k]$$

$$= [0 \ 1 \ 1] \left| \begin{array}{ccccc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right|$$

$$= [\underline{0 \ 1 \ 1} \ \underline{1 \ 0}]_{\text{Systematic}}$$

Info. Parity

Elementary Row Operations (ERO)

□ Elementary row operations are the three fundamental transformations we can apply to generator matrices without changing the fundamental properties of the code they represent.

□ Three Types of Operations:

1. **Row Swapping (Type I):** Exchanging two rows

Symbol: $R_i \leftrightarrow R_j$

Effect: Reorders the basis vectors of the code

2. **Row Scaling (Type II):** Multiplying a row by a non-zero scalar

Symbol: $R_i \rightarrow \alpha R_i$; where $\alpha \neq 0$

Effect: Changes the representation but not the code itself

3. **Row Addition (Type III):** Adding a multiple of one row to another

Symbol: $R_i \rightarrow R_i + \alpha R_j$

Effect: Transforms the basis while preserving the code

Conversion to Systematic Generator Matrix

- ❑ In coding theory, transforming a generator matrix through elementary row operations and column swaps has specific impacts on the resulting linear code

❑ Impact of Elementary Row Operations (EROs)

1. **Preserves the Code:** EROs do not change the code itself. The set of all possible codewords remains exactly the same because the rows still span the same vector space.
2. **Changes the Basis:** It simply provides a different "basis" for the code.
3. **Distance/Weight Properties:** The minimum distance ($d_{\min}$) and weight distribution of the code are completely unaffected.
4. **Encoding Logic:** While the code is the same, the mapping between message bits and codewords changes.

Conversion to Systematic Generator Matrix

□ Impact of Column Swaps

1. **Creates an Equivalent Code:** Column swaps do not preserve the exact code; instead, they create a permutationally equivalent code.
2. **Changes the Codewords:** The individual codewords change because the positions of the bits are rearranged.
3. **Preserves Fundamental Properties:** Even though the codewords look different, the "quality" of the code remains the same:
 - The minimum distance ($d_{\min}$) remains unchanged.
 - The error-correcting capability remains identical.
 - The weight enumerator (how many words have how many 1s) stays the same.
4. **Hardware/Software Impact:** Swapping columns requires you to keep track of the new bit order so that the receiver can "un-swap" them to retrieve the original message.

Linear Block Code Solved

□ Example 4: Determine the parity check Matrix for (7, 4)

Examples

Generator Matrix G :

$$\Rightarrow G = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [i_k : P]$$

$$\rightarrow (7, 4) = (n, k)$$

$$- n = 7$$

$$- k = 4$$

$\rightarrow$ Parity Matrix P

$\rightarrow$ Find P^T

$$[P] = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$[P]^T = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$\rightarrow$ Parity check matrix $H = [P^T : i_{n-k}] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

□ Example 5: Prove that GH^T and

$CH^T = 0$.

$$\rightarrow GH^T = [i_k : p] [p^T : i_{n-k}]^T$$

$$= [i_k : p] \begin{bmatrix} p^T \\ \vdots \\ i_{n-k} \end{bmatrix}$$

mod 2



$$= i_k p \oplus p i_{n-k}$$

$$= p \oplus p$$

$$= 0$$

XOR

$$0 \ 0 \rightarrow 0$$

$$1 \ 1 \rightarrow 0$$

$$\rightarrow CH^T = [i][G][H]^T$$

$$= [i] 0$$

$$= 0$$

□ Example 6: Find the Error Syndrome of received codeword $V = [1\ 1\ 0\ 1\ 1\ 0\ 1]$, Also correct the Error. Here Parity Check Matrix is given by:

$$\Rightarrow H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

∴ Error Syndrome

$$[s] = [V][H]^T$$

$$= [1\ 1\ 0\ 1\ 1\ 0\ 1] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5^{th} \\ = \end{matrix}$$

$$= [1\ 0\ 0]$$

$$\rightarrow e = [0\ 0\ 0\ 0\ 1\ 0\ 0]$$

$$V = [1\ 1\ 0\ 1\ 1\ 0\ 1]$$

$$\therefore x = e \oplus V$$

$$= [1\ 1\ 0\ 1\ 0\ 0\ 1]$$

↑
Correction

- Example 7: If the minimum hamming distance of a linear block code is 3. Find Code error detection and error correction capability.

$$\leadsto d_{\min} = 3$$

$$\begin{aligned}\leadsto \text{Error Detection} &\leq d_{\min} - 1 \\ &\leq 2 \text{ bits}\end{aligned}$$

$$\begin{aligned}\leadsto \text{Error Correction} &\leq \left(\frac{d_{\min} - 1}{2} \right) \\ &\leq 1 \text{ bit}\end{aligned}$$

Linear Block Code Solved

□ Example 8: For a (6, 3) Systematic Linear Block code, the three parity check digits are $C_4 = d_1 + d_2 + d_3$, $C_5 = d_1 + d_2$, and $C_6 = d_1 + d_3$

□ Find,

- I. All Corresponding Code Vectors
- II. Minimum Hamming Distance
- III. Error Detection and Error Correction Capability
- IV. Generator Matrix
- V. Parity Check Matrix
- VI. Find error if received code is [1 0 1 1 0 0]

$$\leadsto (6, 3) = (n, k) \quad \rightarrow \text{Code word}$$

$$- n = 6$$

$$- k = 3$$

$$[c] = [c_1 \ c_2 \ c_3 \ c_4 \ c_5 \ c_6]$$

$$= [i : p_c]$$

$$= [d_1 \ d_2 \ d_3 \ c_4 \ c_5 \ c_6]$$

C	d ₁	d ₂	d ₃	c ₄	c ₅	c ₆	W
c ₁	0	0	0	0	0	0	1
c ₂	0	0	1	1	0	1	3
c ₃	0	1	0	1	1	0	3
c ₄	0	1	1	0	1	1	4
c ₅	1	0	0	1	1	1	4
c ₆	1	0	1	0	1	0	3
c ₇	1	1	0	0	0	1	3
c ₈	1	1	1	1	0	0	4

$$\Rightarrow d_{\min} = W_{\min} = 3$$

$$\Rightarrow \text{Error Detection} \leq d_{\min} - 1$$

$$\leq 2 \text{ bits}$$

$$\Rightarrow \text{Error Correction} \leq \left(\frac{d_{\min} - 1}{2} \right)$$

$$\leq 1 \text{ bit}$$

$$\Rightarrow P_c = \begin{vmatrix} d_1 & d_1 & d_1 \\ d_2 & d_2 & 0 \\ d_3 & 0 & d_3 \end{vmatrix} = [i][p] \Rightarrow \text{Parity Matrix } P = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix}$$

~ Generator Matrix

$$G = [i_k : p] = \begin{vmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{vmatrix}$$

~ Parity Check Matrix

$$H = [p^T : i_{n-k}] = \begin{vmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{vmatrix}$$

~ Error Syndrome

$$S = [y][H^T] = [101100] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [110]$$

¹
2nd bit error

$$\sim e = [0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$y = [1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0]$$

$$\sim x = e \oplus y$$

$$= [1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0]$$

↑
Correction